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T11 – 05

Assignment No. 4

AIM: Write a typescript program to implement inheritance for following example. Bicycle is the base class and MountainBike is derived class. Define arrow functions apply Brake() and speedUp) to decrease and increase speed of bicycle. Function toString() is to display gear and speed. Perform object initialization by using constructor.

LO2: Understand Basic Concepts of Typescript for designing web applications

PROGRAM:

```
// Base class Bicycle
class Bicycle {
  gear:
  number;
  speed: number;

  constructor(gear: number, speed:
  number) {
    this.gear = gear;
    this.speed = speed;
  }

  // Arrow function to apply brake and decrease speed
  applyBrake = (): void => {
    this.speed = Math.max(0,
    this.speed - 5);
    console.log(`Brake applied! Current speed:
    ${this.speed} km/h`);
  }
}
```

```
};
```

```
// Arrow function to increase speed    speedUp = (): void =>
```

```
{    this.speed += 5;    console.log(`Speed increased!
```

```
Current speed:
```

```
${this.speed} km/h`);
```

```
};
```

```
// Method to display gear and speed    toString = (): string
```

```
=> {    return `Bicycle Gear: ${this.gear}, Speed: ${this.speed}
```

```
km/h`;
```

```
};
```

```
}
```

```
// Derived class MountainBike, inheriting from Bicycle    class
```

```
MountainBike extends Bicycle {    suspensionType: string;
```

```
constructor(gear: number, speed: number, suspensionType:
```

```
string) {    super(gear, speed);    this.suspensionType =
```

```
suspensionType;
```

```
}
```

```
// Overriding toString method to include suspensionType

toString = (): string => {

    return `MountainBike Gear: ${this.gear}, Speed: ${this.speed} km/h,
Suspension: ${this.suspensionType}`;

};

}

const myBicycle = new Bicycle(3, 10);

console.log(myBicycle.toString());

myBicycle.speedUp(); myBicycle.applyBrake();


const myMountainBike = new MountainBike(5, 15, 'Full
Suspension');      console.log(myMountainBike.toString());

myMountainBike.speedUp(); myMountainBike.applyBrake();
```

OUTPUT:

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v5.7.3RunExportShare

```
1 // Base class Bicycle
2
3 class Bicycle {
4
5   // Properties of Bicycle class
6
7   speed: number;
8   gear: number;
9
10
11
12
13   // Constructor to initialize speed and gear
14   constructor(speed: number, gear: number) {
15
16     this.speed = speed;
17
18     this.gear = gear;
19
20   }
21
22
23
24
25   // Arrow function for applying brake to decrease speed
26   applyBrake = () => {
27
28     this.speed -= 1; // Decrease speed by 1 unit
29
30
```

JS.DTSErrorsLogsPlugins

[LOG]: "Gear: 3, Speed: 10"

[LOG]: "Speed after speeding up: 11"

[LOG]: "Speed after applying brake: 10"

[LOG]: "MountainBike - Gear: 5, Speed: 15, Suspension: Full Suspension"

[LOG]: "Speed after speeding up: 16"

[LOG]: "Speed after applying brake: 15"